

Mathematical Statistics

Lecture 1: Introduction and Course Overview Probability and Statistics

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1 Introduction and Motivation

1.1 Why Study Probability & Statistics?

Why Study Probability & Statistics?

This semester lectures explores Probability and Statistics, two of the most powerful mathematical tools for understanding and describing the complex world around us.

- **Foundational Importance:** Alongside Calculus, Linear Algebra, and Differential Equations, Probability and Statistics form the bedrock of modern quantitative analysis.
- **Relevance in the Modern Era:** These tools are absolutely essential for understanding data science, machine learning, and artificial intelligence. They provide the mathematical language to handle data and uncertainty.

1.2 The Core Concept: Modeling Uncertainty

The central theme of this course is **uncertainty**. The real world is filled with systems that are too complex to measure or model perfectly.

Key Idea: The Role of Uncertainty

When a system is too complex to measure every detail (like the position of every air molecule in a room) or to model every physical interaction perfectly, we use probability to create an effective and predictive model. We are not saying the system is *truly* random, but that from our perspective as observers with limited information, it is best described probabilistically.

2 Real-World Applications of Probability

Examples are the best way to understand the power of this field. Here are several domains where probability is indispensable.

2.1 Example 1: From Gas Molecules to Thermodynamics

- **The System:** A mole of gas contains an immense number of molecules ($\sim 10^{23}$).
- **The Challenge:** It is impossible to track the position and velocity of every single molecule.
- **The Probabilistic Solution:** Instead of tracking individual molecules, we describe the entire system using a few statistical quantities that emerge from the collective behavior: **Temperature, Entropy, Density**, etc.

2.2 Example 2: Measurement Error

- **The Concept:** If you perform an experiment multiple times (e.g., measuring the acceleration of gravity, g), you will get slightly different results each time.
- **The Probabilistic Model:** This measurement error is not arbitrary. It often follows a predictable pattern, the **Normal (or Gaussian) Distribution**.

- **Historical Context:** The great mathematician Pierre-Simon Laplace was a pioneer in this area, developing much of the theory to quantify the uncertainty in physical measurements.

2.3 Example 3: Control Systems and the Kalman Filter

- **The System:** A dynamic system we want to control, like a self-driving car.
- **The Challenge:** The system is subject to uncertainties from sensor errors and unmodeled forces (wind, rain, hills).
- **The Probabilistic Solution:** The **Kalman Filter** is a brilliant algorithm that merges the system's dynamics with a probabilistic model of these uncertainties to make an optimal estimate of the system's true state.

◎ Key Idea: Deterministic Chaos vs. True Randomness

Many systems we model with probability, like a **coin flip**, are technically deterministic, governed by the laws of physics. However, the outcome is extremely sensitive to tiny, unmeasurable changes in initial conditions. For all practical purposes, the outcome is unpredictable and best modeled as a random event.

Conclusion: Probability is our tool for modeling systems where perfect prediction is beyond our grasp, whether due to fundamental randomness or practical limitations.

3 The Fundamental Difference: Probability vs. Statistics

These two fields are deeply connected but address opposite questions. Think of them as two sides of the same coin.

Probability vs. Statistics

Probability	Statistics
What You Have: A known model (a known probability distribution). For example, "This is a fair coin, so $P(\text{Heads}) = 0.5$ ".	What You Have: Known data (a set of observations). For example, "I flipped a coin 100 times and got 62 heads".
What You Want: To predict the likelihood of future, unknown data . For example, "What is the chance of getting ≥ 7 heads in the next 10 flips?"	What You Want: To infer the properties of the unknown model that generated the data. For example, "Is this a fair coin?"
The Question: What is the probability of observing this data, <i>given</i> that we know the model? $P(\text{Data} \text{Model})$	The Question: What is the most probable model, <i>given</i> the data we have observed? $P(\text{Model} \text{Data})$

We will begin with **Probability** to build the mathematical foundation of models we will later use in **Statistics**.

4 Course Outline - Part I: Probability

4.1 Foundations: Intuition and Counting

- **Focus:** Building intuition through classic examples (coins, cards, dice).
- **Core Concept:** At its heart, probability is an advanced and systematic way of **counting**.

4.2 The Language of Probability: Random Variables & Distributions

- **Random Variable (X):** A variable whose value is a numerical outcome of a random phenomenon.
- **Probability Distribution:** A function that describes the likelihood of a random variable taking on a certain value, often written as $P(X = v|\theta)$.

A Zoo of Important Distributions We Will Study:

Distribution	Description	Example
Bernoulli	A single trial with two outcomes.	One coin flip.
Binomial	Number of successes in n trials.	Number of heads in 10 flips.
Normal (Gaussian)	The classic "bell curve".	Measurement error, heights.
Poisson	Number of rare events in an interval.	Emails received in an hour.
Exponential	Waiting time between rare events.	Time until the next call arrives.

4.3 Describing Distributions: Expected Value, Variance, etc.

We summarize a probability distribution with a few key numbers:

- **Expected Value (Mean):** The long-run average value of the random variable.
- **Variance (and Standard Deviation):** A measure of how "spread out" the distribution is.

4.4 The Cornerstone: The Central Limit Theorem (CLT)

🕒 Key Idea: The Central Limit Theorem (CLT)

The CLT states that if you take a large number of independent and identically distributed random variables and average them, the distribution of that **average** will be approximately **Normal (Gaussian)**, *regardless of the original distribution of the variables*.

In simple terms: Sums and averages of random things tend to follow a bell curve.

Mathematical Notation: If you have random variables $X_1, X_2, \dots, X_n$, their sample mean is

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

The CLT states that as n gets large, the distribution of $\bar{X}$ approaches a Normal distribution.

5 A Glimpse into Part II: Statistics

Once we have the tools of probability, we'll flip the problem around.

- **The Goal:** Use data to make inferences about the world.
- **The Question:** What is the most likely model (θ) that produced the data we observed? $P(\theta|\text{Data})$
- **Topics to be Covered:**
 - **Hypothesis Testing:** Is a new drug effective?
 - **Survey Sampling:** Using a small poll to understand a large population.
 - **Parameter Estimation & Machine Learning:** "Training" a model is finding the parameter values θ that best explain the data.
 - **Bayesian Statistics:** A framework to update our beliefs by combining prior knowledge with new data.

6 Summary of Lecture 1

≡ Lecture Summary

1. **Probability & Statistics** are essential tools for describing the uncertainty and complexity of the real world, with critical applications in science, engineering, and machine learning.
2. **Probability** starts with a **known model** and predicts the likelihood of future data.
3. **Statistics** starts with **known data** and tries to infer the properties of the unknown model that generated it.
4. The first part of the course will build the foundational tools of probability, including **random variables**, key **probability distributions**, and the incredibly important **Central Limit Theorem**.
5. This foundation will directly enable us to tackle powerful statistical methods in the second half of the course.